

EXPERIMENT – 3

❖ Aim :

To design and develop an **Interactive Quiz Web Application** using **HTML, CSS, and JavaScript**.

❖ Tasks :

- **Project Setup & Introduction** – Create the project folder and link HTML, CSS, and JavaScript files properly.
 - **Quiz Interface Design** – Design quiz layout with questions, options, and navigation buttons using semantic HTML.
 - **Styling with CSS** – Apply colors, spacing, alignment, and layout to improve user experience.
 - **Quiz Logic Implementation** – Store questions and answers in JavaScript and handle user input.
 - **Score Tracking & Navigation** – Move between questions and calculate the score dynamically.
-

❖ Tech Stack Used :

- **HTML** • **CSS** • **JavaScript**
-

❖ Html Code :

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <title>Interactive Quiz</title>

    <link rel="stylesheet" href="style.css">

</head>

<body>

<header>

    <h1>Interactive Quiz</h1>

    <p>Test your Web Development knowledge</p>
```

```
</header>
```

```
<main class="quiz-container">
```

```
  <section id="intro">
```

```
    <h2>Quiz Introduction</h2>
```

```
    <p>This quiz contains multiple-choice questions. Click Start to begin.</p>
```

```
    <button onclick="startQuiz()">Start Quiz</button>
```

```
  </section>
```

```
  <section id="quiz" class="hidden">
```

```
    <h2 id="question"></h2>
```

```
    <div id="options"></div>
```

```
    <button onclick="nextQuestion()">Next</button>
```

```
  </section>
```

```
  <section id="result" class="hidden">
```

```
    <h2>Your Result</h2>
```

```
    <p id="scoreText"></p>
```

```
    <p id="feedback"></p>
```

```
    <button onclick="restartQuiz()">Restart Quiz</button>
```

```
  </section>
```

```
</main>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

► **CSS CODE :**

```
* {  
    box-sizing: border-box;  
    font-family: Arial, sans-serif;  
}
```

```
body {  
    margin: 0;  
    background: linear-gradient(to right, #74ebd5, #9face6);  
}
```

```
header {  
    background-color:blueviolet ;  
    color: white;  
    padding: 20px;  
    text-align: center;  
}
```

```
.quiz-container {  
    background-color: white;  
    max-width: 600px;  
    margin: 30px auto;  
    padding: 25px;  
    border-radius: 10px;  
    box-shadow: 0 10px 25px rgba(0,0,0,0.2);  
}
```

```
button {  
    background-color: #3498db;  
    color: white;
```

```
border: none;
padding: 10px 18px;
margin-top: 15px;
cursor: pointer;
border-radius: 5px;
}
```

```
button:hover {
    background-color: #2980b9;
}
```

```
.option {
    background-color: #ecf0f1;
    padding: 10px;
    margin: 8px 0;
    border-radius: 5px;
    cursor: pointer;
}
```

```
.option:hover {
    background-color: #dcdde1;
}
```

```
.hidden {
    display: none;
}
```

```
footer {
    background-color: #2c3e50;
```

```
color: white;
text-align: center;
padding: 10px;
}
```

► JS CODE :

```
const quizData = [
  {
    question: "What does HTML stand for?",
    options: [
      "Hyper Text Markup Language",
      "High Text Machine Language",
      "Hyperlinks Text Mark Language",
      "None of these"
    ],
    correct: 0
  },
  {
    question: "Which language is used for styling web pages?",
    options: ["HTML", "CSS", "Java", "Python"],
    correct: 1
  },
  {
    question: "Which is used for web page interactivity?",
    options: ["HTML", "CSS", "JavaScript", "SQL"],
    correct: 2
  }
];
```

```
let currentQuestion = 0;
```

```
let score = 0;
```

```
function startQuiz() {
```

```
    document.getElementById("intro").classList.add("hidden");
```

```
    document.getElementById("quiz").classList.remove("hidden");
```

```
    loadQuestion();
```

```
}
```

```
function loadQuestion() {
```

```
    const q = quizData[currentQuestion];
```

```
    document.getElementById("question").innerText = q.question;
```

```
    const optionsDiv = document.getElementById("options");
```

```
    optionsDiv.innerHTML = "";
```

```
    q.options.forEach((option, index) => {
```

```
        const div = document.createElement("div");
```

```
        div.className = "option";
```

```
        div.innerText = option;
```

```
        div.onclick = () => selectAnswer(index);
```

```
        optionsDiv.appendChild(div);
```

```
    });
```

```
}
```

```
function selectAnswer(selected) {
```

```
    if (selected === quizData[currentQuestion].correct) {
```

```
        score++;
```

```
    }
```

```

    nextQuestion();
}

function nextQuestion() {
    currentQuestion++;
    if (currentQuestion < quizData.length) {
        loadQuestion();
    } else {
        showResult();
    }
}

function showResult() {
    document.getElementById("quiz").classList.add("hidden");
    document.getElementById("result").classList.remove("hidden");

    document.getElementById("scoreText").innerText =
        `Your Score: ${score} / ${quizData.length}`;
    let feedback = "Try Again";
    if (score === quizData.length) feedback = "Excellent";
    else if (score >= 2) feedback = "Good";
    document.getElementById("feedback").innerText = feedback;
}

function restartQuiz() {
    currentQuestion = 0;
    score = 0;
    document.getElementById("result").classList.add("hidden");
    document.getElementById("intro").classList.remove("hidden")
}

```

► Output :

Interactive Quiz

Test your Web Development knowledge

Quiz Introduction

This quiz contains multiple-choice questions. Click Start to begin.

Start Quiz

Interactive Quiz

Test your Web Development knowledge

What does HTML stand for?

Hyper Text Markup Language

High Text Machine Language

Hyperlinks Text Mark Language

None of these

Next

Interactive Quiz

Test your Web Development knowledge

Your Result

Your Score: 3 / 3

Excellent

Restart Quiz